



The Maths of Predators and Preys

Don't get Eaten by the Wolf

Hares (No Predators)



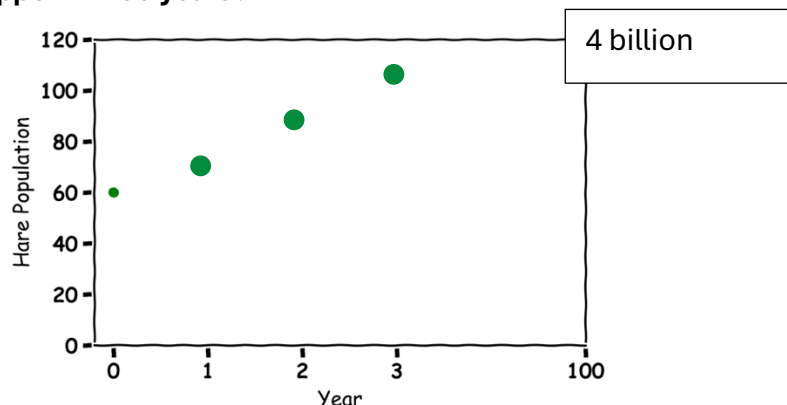
Given a population of 60 Hares and a growth rate of 0.2 what would the population be after three years?

$$H_{\text{Future}} = H + 0.2(H)$$

Year	Hare Population
0	60
1	$60 + 0.2 \cdot (60) = 72$
2	$72 + 0.2 \cdot (72) = 86$
3	$86 + 0.2 \cdot (86) = 103$

Sketch the Graph.

Guess what would happen in 100 years?



Lynx (No Prey)



Given a population of 30 Lynxes and a death rate of 0.1 what would the population be after three years?

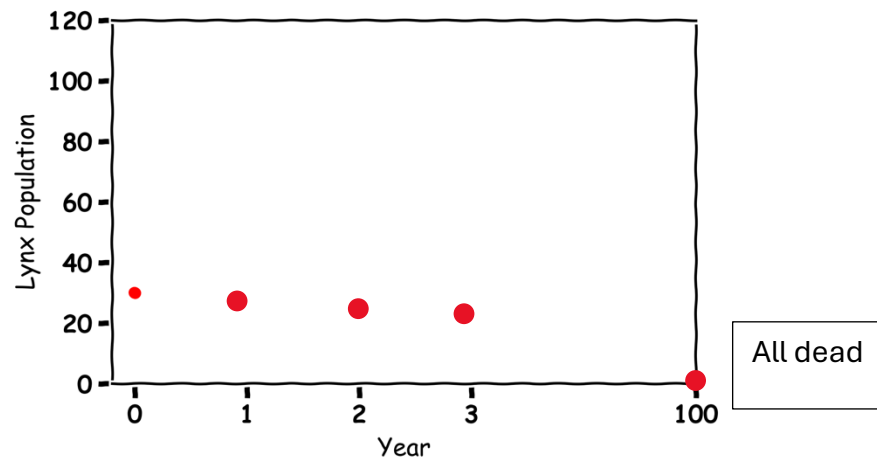
$$L_{\text{Future}} = L - 0.1(L)$$

Year	Lynx Population
0	30
1	$30 - 0.1 \cdot (30) = 27$
2	$27 - 0.1 \cdot (27) = 24$
3	$24 - 0.1 \cdot (24) = 22$



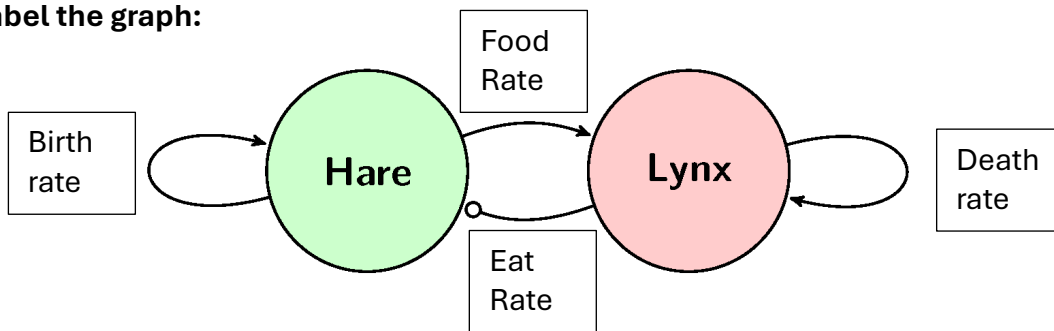
Sketch the Graph.

Guess what would happen in 100 years?



Hares and Lynxes

Label the graph:



Given a population of 60 Hares and 30 Lynxes with a birth rate of .2, a death rate of 0.1, a eat rate of 0.005 and a food rate of 0.002 what would the populations be after three years?

	Value		Value
Birth	0.2	Death	0.1
Eat	0.005	Food	0.002

$$H_{\text{Future}} = H + 0.2(H) - 0.005(H)(L)$$

$$L_{\text{Future}} = L - 0.1(L) + 0.002(L)(H)$$

Year	Hare Population	Lynx Population
0	60	30
1	$60 + 0.2 \cdot (60) - 0.005 \cdot (60) \cdot (30) = 63$	$30 - 0.1 \cdot (30) + 0.002 \cdot (30) \cdot (60) = 31$
2	$63 + 0.2 \cdot (63) - 0.005 \cdot (63) \cdot (31) = 66$	$31 - 0.1 \cdot (31) + 0.002 \cdot (31) \cdot (63) = 31$
3	$66 + 0.2 \cdot (66) - 0.005 \cdot (66) \cdot (31) = 69$	$31 - 0.1 \cdot (31) + 0.002 \cdot (31) \cdot (66) = 32$